Chemin Ahn

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Birth: 1998.12.22
Portfolio: https://chemx3937.github.io/

Education

- Bachelor of Science in Mechanical Engineering
  Chungang University, Seoul, South Korea
  - Major in Mechanical Engineering
  - 3.84 / 4.50 GPA (CUM LAUDE)
- Master Course of Science in Mechanical Engineering
  Sungkyunkwan University, Suwon, South Korea
  - Major in Mechanical Engineering
  – Robotics Innovatory Lab (Prof. Hyouck Ryeol Choi)

Research Interest

- Robotics
- Imitation Learning
- Teleoperation
- Compliance Control
- VLA
- Reinforcement Learning
- Computer Vision

Projects

Master Course

- **Teleoperation:**
  - Single Arm Leader Arm
    w/ Teleoperation + Force Feedback + Gripper Feedback + Gravity Compensation
    (https://youtu.be/dhNTIGTX53Q?si=il2BSdpKIuVkS4Q8)
  - Dual Arm Leader arm
    w/ Teleoperation + Force Feedback + Gravity Compensation
    (https://youtu.be/Q0x2Fh5BNF8?si=XVNefWEzQ33wNWMp)
  - RB Y-1 Master Arm Force Feedback Teleoperation
    (https://youtu.be/adola9Bo_tA?si=rqvpB0QddU2BJhsM)
  - Using VR Tracker
    (https://youtu.be/ToG8sNmdli0?si=a3Xqx6uPEkkllsuQ)
  - Using Visual SLAM based VR Tracker
    (https://youtu.be/yQAdfKhJR1M?si=dOepBiyha9SJi26N)

- **Imitation Learning**
  - Dual Arms + Hands (https://youtu.be/Vs-ICyzwaLc)
  - Dual Arms (https://youtu.be/WTx4ySlACAU)
  - One Arm (https://youtu.be/VPG-_bH2SSY)
  - Multimodal (Vision + Force) Imitation Learning – On going

Intern & Others

- Doosan Robotics Robot Control Team intern: 2024.07.01~10.04
  - Project: Visual Servoing Package (https://youtu.be/9UFAdaa_PR8?si=LtkzL9kk2eVBcLp8)
- AI-Robotics Academy (KG-Kairos): 2023.12.19~24.06.17
  - Project: assisting pharmaceutical preparation and autonomous delivery using cobot and AMR (https://youtu.be/-mTXpCWZacI?si=YXC8AwRQ8FV_uN9S)

Bachelor's degree

- Finding the optimal sensor for AMR and analyzing driving characteristics by algorithm combinations (https://youtu.be/whuUySDWIwA?si=vPVX8IQul7cd9p3q)
- Undergraduate research student: 2022.12.22~2023.02.28
  - Design light weight passive wearable suit for logistic workers
- Design light weight battle robot
- Maze escape using OD based autonomous drone

Awards

- KRoC 2026 RED Show @ KROS(KRoC 2026) – 1st prize (2026.02)
- Capstone Design Contest @ Chungang University – 2nd prize (2024.06)
- AI-Robotics Academy @ KG-Kairos (KG ICT) – 2nd prize (2024.06)
- Cum Laude @ Chungang University (2025.02)

Patent

- Force Feedback Leader arm with Gravity Compensation - Ongoing

Imitation Learning Policies

- ACT
- Diffusion Policy
- DiT-Policy

Tool Skills

- Language & Framework: Python, ROS 1, 2, MATLAB
- Libraries: Pytorch, TensorFlow, OpenCV, Numpy, Pandas, matplotlib
- Simulation Tools: Isaac Sim, Isaac Lab, Gazebo, Ignition Gazebo
- Manipulator: Doosan Robotics, Rainbow Robotics
- Other: Git, SolidWorks

Language Skills

- English: fluent (reading, listening); intermediate (speaking, writing)
  - TOEIC Speaking: 150(IH)

Additional Link

- Portfolio: https://chemx3937.github.io/
- Gihub: https://github.com/Chemx3937
- Youtube: https://www.youtube.com/@%EC%B3%84-r3i
- Linkedin: https://www.linkedin.com/in/chem1222/

Related Courses

- Intelligent Robotics
- Robot Reinforcement Learning
- Modern Control Systems
- Human-Robot Collaboration
- Dynamics
- System Dynamics
- Control Systems Engineering
- Mechatronics
- Robotics
- Numerical Methods
- Design and Analysis of Experiments
- System Control Experiments
- Visual Programming